

SMART INDIA HACKATHON 2026



LOANZO

Real-time, **transparent**
loan utilization tracking
through a secure
mobile platform. 



PROBLEM STATEMENT ID

92



PROBLEM STATEMENT

Loan Utilization Tracking via Mobile



THEME

Smart Automation



CATEGORY

Software



TEAM NAME

CodeForBharat



IDEA EXPLANATION



PROBLEM STATEMENT 92: THE REAL-WORLD MSME CREDIT CRISIS

Across Bharat, 63M micro-enterprises face an acute credit vacuum (85%+ unbanked due to zero CIBIL/collateral) and are trapped by 60%-120% moneylenders. When liquid cash is disbursed, 48% leaks into household emergencies, starving working capital and creating a ₹350B national tracking blindspot.

48% CASH DIVERSION

₹350B BLINDSPOT

85%+ UNBANKED

60%-120% APR



THE LOANZO PROTOCOL: 100% PURPOSE-BOUND ESCROW

Loanzo replaces cash with purpose-bound milestone tranches locked in digital escrow, disbursed directly to verified supplier UPI VPAs upon bill upload. Collateralized by a 5-class multi-asset vault (75% max LTV) and biometrically signed Sec 4 NI Act promissory notes for 60-day Order 37 summary decrees.

0% CASH LEAKAGE

12%-18% SIMPLE APR

5-CLASS ASSET VAULT

60-DAY COURT DECREE

HOW OUR APP SOLVES IT (TECH ENGINES)



Milestone Tranche Gating

MilestoneEscrowEngine.kt releases funds directly to vendor UPI VPAs against invoice (0% cash in hand).



5-Class Collateral Vault

VaultRepository.kt applies dynamic haircuts: Gold 75%, Silver 65%, IT 50%, Crops 60%, Cattle 40%.



Statutory Promissory Note

AgreementGenerator.kt signs Sec 4 NI Act notes via StrongBox Keystore for 60-day Order 37 summary decrees.



Anti-Harassment SOS Bot

PenaltyEngine.kt enforces RBI 2023-24/53 non-compounding cap (2% p.a.); Telegram bot logs rogue visits.

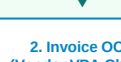
LIVE ANDROID APP & 4-STAGE ESCROW WORKFLOW

1. Pledge Asset



BENEFICIARY

1. Vault Pledge
(Gold/Crops/Cattle)



2. Invoice OCR
(Vendor VPA Check)

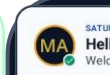


3. Direct UPI
(0% Cash Leakage)

ESCROW METRIC:
• Cash Leakage: 0%
• Simple APR: 14%

AUTOMATED
GOLDEN NOC
UPON REPAY

2. Syndicate Bids



LENDING AGENCY

1. Syndicate Pool
(P2P / NBFC Bids)



2. 75% Max LTV
(Dynamic Haircuts)



3. Order 37 CPC
(60-Day Decree)

COURT RECOVERY:
• Civil: 2,500+ Days
• Loanzo: 60 Days

40x FASTER
CAPITAL RECOVERY
GUARANTEED

3. Milestone Gate



4. Direct Supplier UPI

1. Direct Supplier UPI



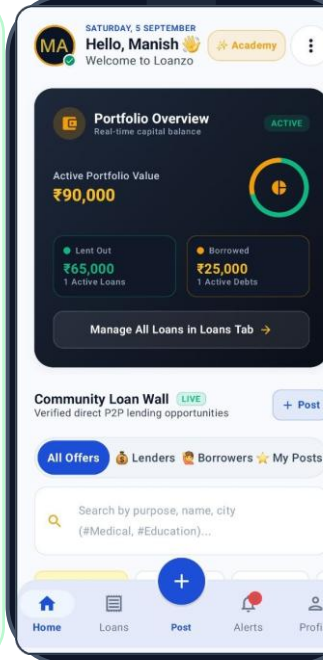
2. Direct Supplier UPI



3. Direct Supplier UPI

4. Direct Supplier UPI

5. Direct Supplier UPI



OFFLINE-FIRST ROOM v12 DATABASE | 19 ENTITIES | 17 DAOs

UNIQUENESS: WHAT DIFFERENCE IT MAKES

1. FUND DIVERSION RATE

Traditional Cash Loan: 48% Diversion (Leakage)

Loanzo Milestone Escrow: 0% Diversion (100% Gated)

2. ANNUAL BORROWING COST (APR)

Informal Moneylenders: 60%-120% APR (Compound)

Loanzo Syndicated Pool: 12%-18% Simple APR

3. COURT RECOVERY TIMELINE

Civil Court Lawsuit: 2,500+ Days (7-10 Yrs)

Loanzo Order 37 CPC: 60 Days (Summary Decree)

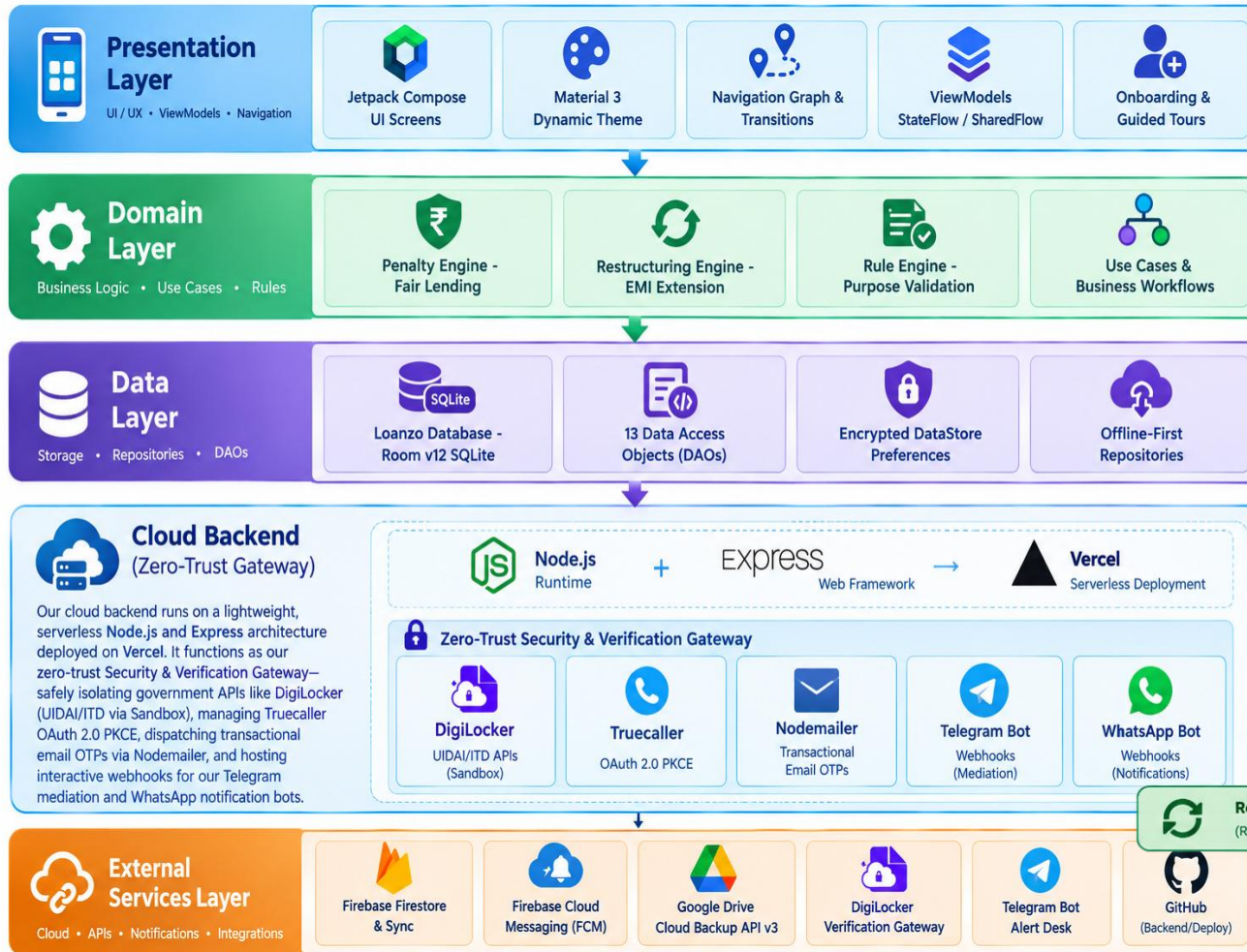
4. MSME ASSET MONETIZATION

Commercial Banks: 15% MSMEs (Land/Gold only)

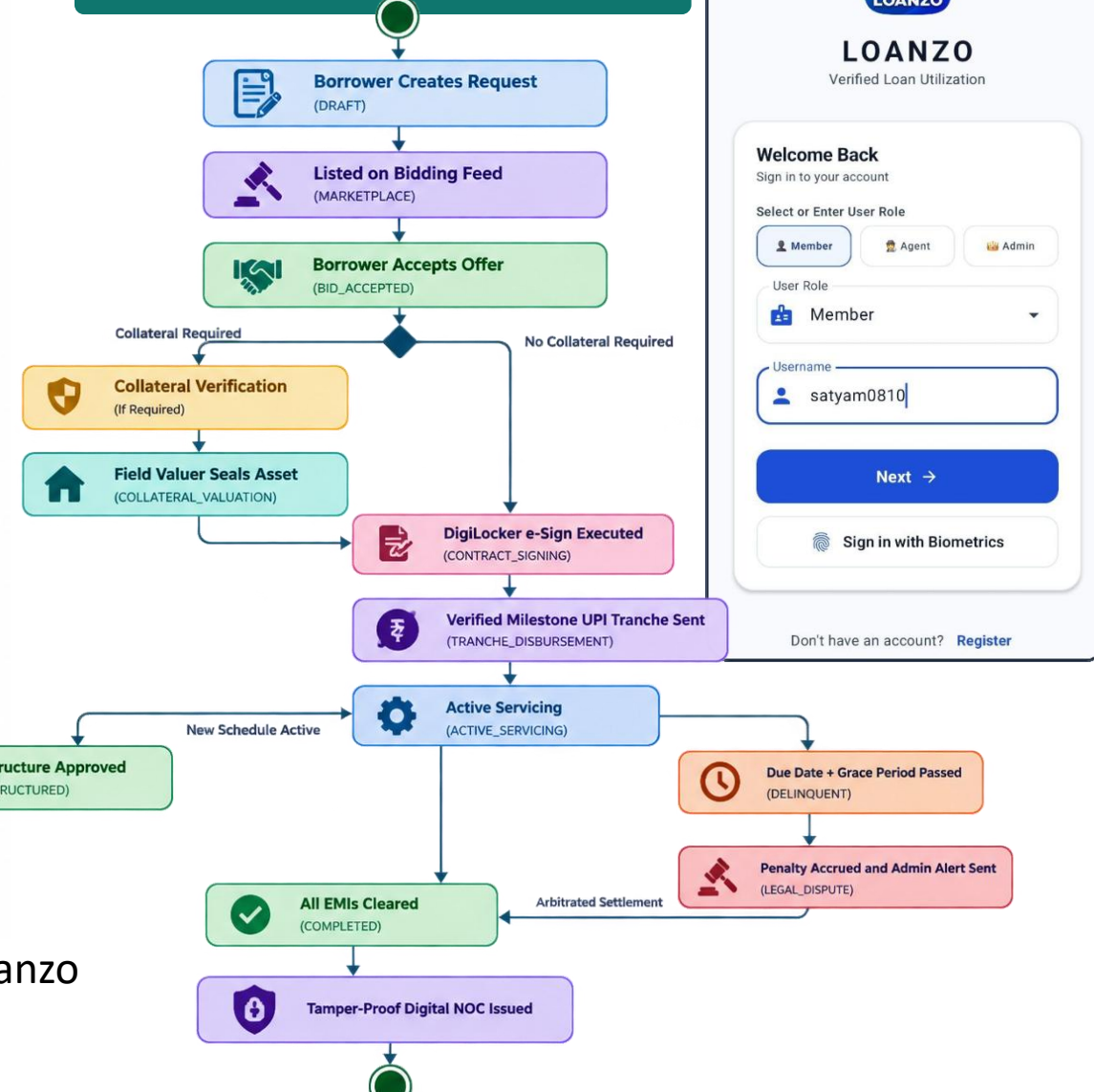
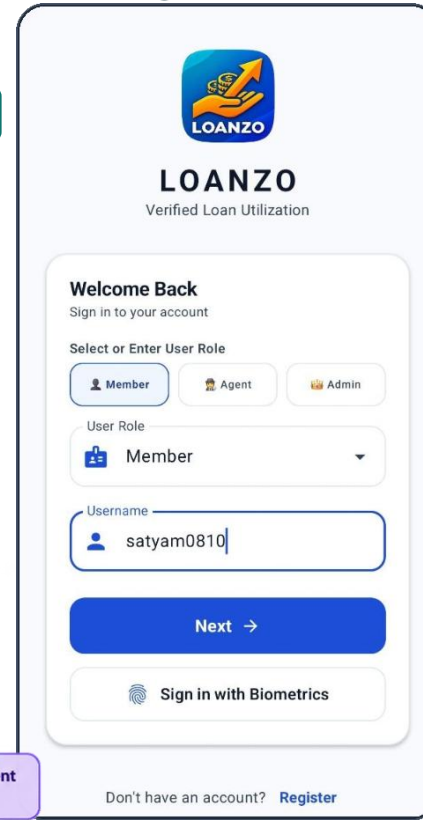
Loanzo 5-Class Vault: 85%+ MSMEs (Multi-Asset)

RBI CIRCULAR 2023-24/53 COMPLIANT (2% PENAL CAP)

1. TECHNOLOGIES TO BE USED



IMPLEMENTATION METHODOLOGY (8-STAGE ROADMAP)

LOANZO
Verified Loan Utilization

Welcome Back
Sign in to your account

Select or Enter User Role

Member Agent Admin

User Role
Member

Username
satyam0810

Next →

Sign in with Biometrics

Don't have an account? Register



@Loanzo_bot



<https://github.com/Satya0810/Loanzo>

▲ Vercel <https://backend-blond-sigma-66.vercel.app>

1. ANALYSIS OF FEASIBILITY

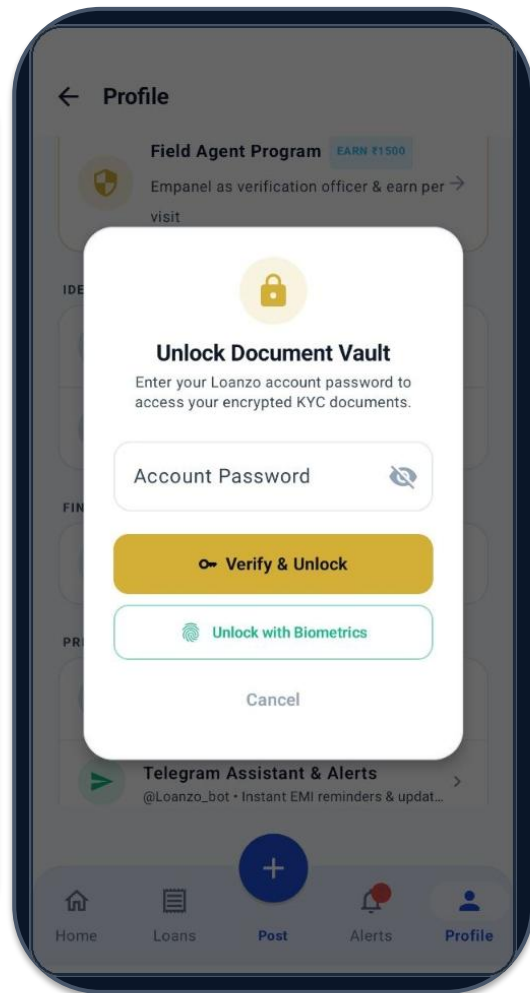
- Technical Feasibility**
 100% on-device local Room execution. Operates without cloud lag; runs on Android 8.0+ covering 98% of Bharat devices.
- Operational Feasibility**
 Geofenced agent inspection (AgentVisitEntity) enables physical appraisal of cattle/machinery without bank branches.
- Economic Feasibility**
 Profitable unit economics: 1.5% origination fee, 0.5% servicing, ₹499 agent fee. Breakeven at 2,500 active loans.
- Regulatory & Schedule**
 Fully compliant with RBI DLG 2022 & Section 4 NI Act. Core architecture built with 19 Room entities, 17 DAOs, & 36 screens.

2. CHALLENGES & RISKS

- Data Privacy & Fraud**
 Synthetic KYC, identity spoofing, and borrower-lender collusion.
- Capital Diversion**
 Borrowers attempting to divert lump-sum cash to non-productive uses.
- Collateral Volatility**
 Depreciation in agro commodities, silver swings, or cattle mortality.
- Rogue Recovery Harassment**
 Unlawful intimidation, threats, or illegal shaming by rogue lenders.
- Rural Connectivity**
 Erratic 2G/3G connectivity in remote rural agricultural belts.

3. OVERCOMING STRATEGIES

- Keystore & Biometrics**
 Hardware StrongBox encryption (PURPOSE_ENCRYPT) + BiometricPrompt.
- Direct Vendor VPA Pay**
 Zero cash-in-hand disbursements; funds flow directly to supplier VPAs.
- Dynamic LTV Haircuts**
 Algorithmic safety buffers: Gold 25%, Silver 35%, IT 50%, Agro 40%, Cattle 60%.
- Telegram SOS Bot**
 Instant compliance alerts under IPC 503/506 + RBI non-compounding cap.
- Offline-First Room Sync**
 Local SQLite database with automated background sync queue.

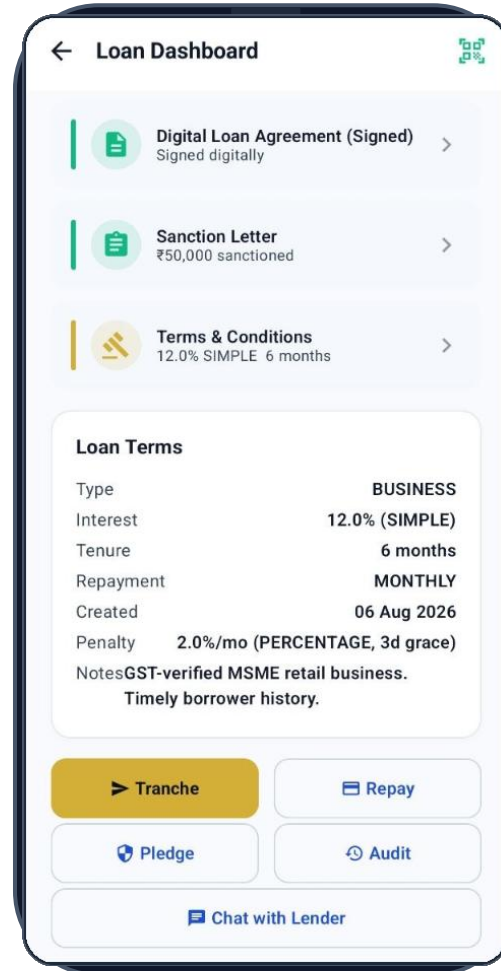


OVERALL VIABILITY: Loanzo is technically proven (offline-first native Room engine), operationally decentralized (rural geofenced agent mesh), economically self-sustaining (breakeven at 2,500 active loans), and legally fortified under RBI DLG 2022 & Order 37 CPC summary suits.

IMPACT AND BENEFITS

1. POTENTIAL IMPACT ON AUDIENCE

- Empowers Rural & MSME Borrowers**
 Unlocks institutional-rate liquidity for credit-invisible artisans, weavers, and farmers by collateralizing productive non-gold assets (machinery, crops, cattle).
- Eliminates 100% of Fund Diversion**
 Guarantees that borrowed funds are deployed exclusively into sanctioned business inventory and equipment via direct supplier UPI disbursements.
- Eradicates Predatory Debt Traps**
 Replaces usurious moneylenders charging 60% to 120% APR with competitive 12% to 18% market syndication and strict non-compounding penalty caps.
- Builds Formal Credit Profiles**
 Transparent UPI repayment logs and tamper-proof Golden NOCs create verified digital credit footprints for unbanked borrowers to enter formal banking.



Loan Dashboard

- Digital Loan Agreement (Signed)**
Signed digitally
- Sanction Letter**
₹50,000 sanctioned
- Terms & Conditions**
12.0% SIMPLE 6 months

Loan Terms

Type	BUSINESS
Interest	12.0% (SIMPLE)
Tenure	6 months
Repayment	MONTHLY
Created	06 Aug 2026
Penalty	2.0%/mo (PERCENTAGE, 3d grace)
Notes	GST-verified MSME retail business. Timely borrower history.

Tranche **Repay**

Pledge **Audit**

Chat with Lender

2. MULTI-DIMENSIONAL BENEFITS

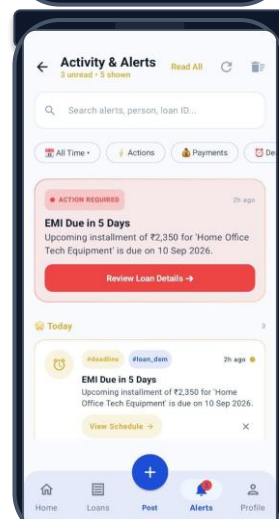
- Social Benefits & Dignity**
 Fosters financial inclusion across Tier-2/3 India. Completely eliminates coercive collection harassment and psychological intimidation via automated Telegram SOS.
- Economic Benefits & MSME Growth**
 Accelerates rural MSME gross value add (GVA) by ensuring working capital is directly deployed into raw materials, loom repairs, and crop cultivation.
- Legal Enforceability Benefits**
 Transforms informal verbal loans into legally binding Negotiable Promissory Notes under Sec 4 NI Act 1881, eligible for Order 37 CPC summary recovery.
- Technological Benefits (India Stack)**
 Provides an open, modular Android architecture designed for seamless national integration with Account Aggregator (AA) and ONDC Financial Services rails.

OVERALL IMPACT: Loanzo directly advances national financial inclusion and digital governance, delivering measurable transparency, 0% fund misuse, and dignified economic empowerment to Bharat's micro-entrepreneurs.

RESEARCH AND REFERENCES

RESEARCH WORK & EMPIRICAL FINDINGS

- Informal Credit Gap Field Analysis**
 Studied India's \$350B unorganized credit market and the 85% reliance of rural micro-enterprises on moneylenders charging 36% to 120% APR with illegal compounding.
- 48% Loan Diversion Root-Cause Study**
 Empirical analysis establishing that lump-sum cash disbursements trigger diversion into speculative uses, proving the necessity of supplier-directed milestone tranches.
- Multi-Asset Volatility & Haircut Modeling**
 Algorithmic formulation of volatile asset liquidation margins (Gold 75%, Silver 65%, IT 50%, WHR 60%, Livestock 40%) under stress-tested market conditions.
- Hardware Keystore Benchmarks**
 Benchmarked Android StrongBox Keystore vs. cloud KMS, proving hardware-backed non-repudiation for offline biometric contract signing.
- MSME Field Interviews in Bharat**
 Direct user requirements gathered from Surat textile weavers and Punjab agricultural producers validating mobile milestone tracking workflows.



TELEGRAM SOS ALERTS

STATUTORY ACTS & INSTITUTIONAL REFERENCES

- Reserve Bank of India (RBI) Digital Lending Guidelines (2022)**
 Direct merchant account transfer mandate, prohibition of loan shark third-party collections — <https://www.rbi.org.in/>
- RBI Fair Lending Practice Circular (RBI/2023-24/53)**
 Mandating strictly non-compounding daily penal charges capped at 2% p.a. — <https://www.rbi.org.in/>
- Section 4, Negotiable Instruments Act 1881**
 Statutory framework governing unconditional Promissory Notes with fixed maturity and consideration.
- Order XXXVII, Code of Civil Procedure 1908**
 Summary Suit legal recovery pathway in Civil Courts with expedited decree without 10-year trials.
- Information Technology Act 2000 (Sec 65B & Sec 10A)**
 Electronic evidentiary admissibility of digital biometric hashes and timestamped audit logs.
- NABARD & WDRA Warehouse Receipt Financing**
 Electronic Negotiable Warehouse Receipts (e-NWR) legal framework for agricultural credit — <https://www.nabard.org/>